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SUMMARY

Backend Java engineer focused on **Spring Boot** services, **JVM performance**, and applied AI integration. Comfortable owning a service end-to-end, from API design and database schema through deployment and on-call. Recent work centers on **Retrieval Augmented Generation** pipelines using **Spring AI** and **LangChain4j**, and on modernizing concurrency models with **Java Virtual Threads**. Independent benchmarking and undergraduate research projects on GitHub back up the work below.

EXPERIENCE

Software Engineer July 2024 – October 2025
Health Insurance Care — California, United States (Hybrid)

- **AI document routing agent (primary contribution):** Designed and built an internal agent that classifies and routes incoming cases by retrieving relevant policy passages from a vector index. Implemented the pipeline using **Spring AI** and **LangChain4j** with **PostgreSQL** plus **pgvector** for embeddings, and chose chunk size and retrieval-k after measuring precision against a labeled evaluation set. Reduced average manual routing time from approximately 8 minutes to 2 minutes per case.
- **Service decomposition:** Contributed 2 of 6 **Spring Boot** services in an ongoing migration from a legacy monolith to an event-driven architecture on **Apache Kafka**. Owned the design of the claim intake service end-to-end, including the **REST API** contract, **PostgreSQL** schema, and **Kafka topic** layout for downstream consumers. Participated in design reviews for the broader migration.
- **JVM and query performance:** Profiled production hot paths with **JDK Flight Recorder**, tuned **garbage collection** settings, and removed **N plus 1 query** patterns using **Hibernate fetch joins**. Reduced peak garbage collection pause times from approximately 180 milliseconds to under 20 milliseconds on the main read path.
- **Concurrency modernization:** Migrated a request fan-out workload from **ExecutorService** to **Java Virtual Threads** after benchmarking both with **JMH**. Documented the tradeoffs (carrier-thread pinning behavior, debugging visibility, observability gaps) for the team and wrote the migration playbook used by other services. (See virtualthread-benchmark-lab in Selected Projects.)
- **API security:** Implemented **OAuth 2.0** plus **JWT-based Single Sign-On** across 3 internal applications using **Spring Security**. Partnered with the security team to address findings from the annual penetration test.
- **Testing standards and code review:** Drove the team's adoption of **Testcontainers** for integration tests against real **PostgreSQL**, replacing an in-memory **H2** setup that was hiding PostgreSQL-specific data type and locking behavior. Reviewed peer pull requests on a weekly rotation and authored the team's internal onboarding guide for new **Spring Boot** services.

Web Developer September 2021 – April 2022
Tristate Health Insurance Group — Cincinnati, OH

- **Customer portal:** Contributed **React** components and routing for a member-facing **Single Page Application** serving approximately 10,000 monthly users. Worked across a multi-developer team using **JSX**, **React Router**, and **Tailwind CSS**.
- **State management design choice:** Picked **Redux Toolkit** with **async thunks** for the agent quoting tool after evaluating **Context API** and plain hooks; documented the decision so later contributors understood the tradeoff between boilerplate and predictability for shared state.
- **Authentication flow:** Implemented **OAuth 2.0** sign-in and sign-out for the member portal, including session expiration handling and clearer error states on the login path. Reduced login latency from approximately 2.1 seconds to 1.25 seconds.
- **Frontend performance:** Improved **Lighthouse** performance score from 72 to 94 by adding **code splitting** and route-level **lazy loading** on the critical render path.
- **Testing:** Wrote **Jest** and **React Testing Library** tests covering the highest-traffic user flows; consistently caught regressions before they reached QA.

- **Deployment:** Containerized the portal with **Docker** and deployed to **Google Cloud** behind **Cloudflare** for HTTPS termination and edge caching.

SELECTED PROJECTS

virtualthread-benchmark-lab — *Java, JMH*

JMH benchmark suite comparing traditional **ExecutorService** thread pools against **Java Virtual Threads** across I/O-bound, CPU-bound, and mixed workloads. Surfaces concrete throughput and latency tradeoffs and documents **carrier-thread pinning** behavior under different concurrency patterns.

ai-agent-service — *Java, Spring Boot 3, LangChain4j*

Spring Boot service that exposes a chat endpoint backed by an AI agent. Uses **LangChain4j** for LLM orchestration, **PostgreSQL** with **pgvector** for embedding storage and **Retrieval Augmented Generation**, and **Model Context Protocol (MCP)** tools for the agent to search Java documentation and code examples. **Virtual threads** are enabled at the **Tomcat** layer to handle concurrent retrieval and LLM calls.

Live Facial Emotion Recognition (Undergraduate Research) — *Northern Kentucky University, 2024*

Research conducted under Dr. Junxiu Zhou, Associate Professor of Computer Science at NKU, specializing in machine learning, deep learning, and computer vision. Designed and trained a custom **5-layer Convolutional Neural Network** in **TensorFlow** and **Keras** for real-time recognition of seven facial emotions, trained on the FER2013 dataset (~36,000 labeled 48x48 grayscale images). Applied **Batch Normalization** and **Dropout** for regularization, **SGD with Nesterov momentum** for optimization, and data augmentation (rotation, shift, zoom, horizontal flip) for robustness. Integrated **MediaPipe Face Mesh** for real-time face detection and landmark tracking, achieving 24 frames per second on mobile devices. Improved baseline classification accuracy from 66.61 percent to 71.20 percent through fine-tuning, reaching 73.40 percent under optimal lighting conditions. Aimed at early detection of emotional distress for mental health applications.

TECHNICAL SKILLS

Languages: Java (11, 17, 21), TypeScript, JavaScript, Python, SQL

Backend: Spring Boot, Spring Data JPA, Spring Security, Spring AI, Hibernate, PostgreSQL, Apache Kafka, Redis

Concurrency and JVM: Virtual Threads (Project Loom), CompletableFuture, JDK Flight Recorder, JVM garbage collection tuning, JMH benchmarking

AI and Machine Learning: LangChain4j, pgvector, embedding-based retrieval, Retrieval Augmented Generation, Model Context Protocol (MCP), TensorFlow, Keras, Convolutional Neural Networks, MediaPipe

Testing: JUnit 5, Mockito, Testcontainers, Jest, React Testing Library

Observability: OpenTelemetry, Jaeger, Micrometer

DevOps: Docker, Maven, Gradle, GitHub Actions, AWS, Google Cloud

Frontend: React, Redux Toolkit, React Router, Tailwind CSS

EDUCATION

Northern Kentucky University Graduated May 2025

Bachelor of Science in Computer Science, Minor in Mathematics — GPA 3.7

Awards and Honors:

- International Merit Scholarship — Northern Kentucky University
- EDGE Award — Northern Kentucky University
- Dr. Charles Frank Scholarship — Northern Kentucky University
- Sehnert-Pugh Scholarship — Northern Kentucky University

CERTIFICATIONS

- Oracle Certified Professional, Java SE 11 Developer — Issued February 2026. Exam Score: 86 percent.
- Java Certified Foundations Associate — Issued November 2025.